

ADMM CASE STUDY

Supply Chain & Inventory Management

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I. INTRODUCTION

1.1 Introduction:

Supply Chain & Inventory Management is an analytics solution developed to analyze sales demand, inventory levels, reorder requirements, supplier lead times, purchase orders, and inventory risk.

The available datasets contain information related to sales transactions, inventory levels, suppliers, and purchase orders. These datasets provide information about products, stores, stock levels, reorder levels, supplier lead times, order quantities, and delivery dates.

The objective of this case study is to transform the available supply chain and inventory data into a structured analytical solution that supports:

- Sales and demand analysis
- Product movement analysis
- Inventory level analysis
- Reorder risk analysis
- Supplier lead-time analysis
- Purchase order analysis
- Demand forecasting
- Reorder point calculation
- Safety stock analysis
- Supplier clustering
- Inventory optimization
- SQL-based business analysis
- Business insight generation

1.2 Business Problem & Objectives:

• Business Problem-

Organizations may experience **stockouts and overstocking** when inventory levels are not aligned with demand and replenishment requirements.

Without an integrated analytical solution, it becomes difficult to efficiently identify:

- Fast-moving and slow-moving products
- Products approaching or falling below reorder levels
- Inventory risk across stores

- Product demand patterns
- Future demand requirements
- Appropriate reorder points
- Safety stock requirements
- Supplier lead-time patterns
- Purchase-order performance
- Inventory optimization opportunities

The case study therefore focuses on building an analytical pipeline that converts sales, inventory, supplier, and purchase-order data into structured **business insights for inventory and supply-chain management.**

• Objectives-

The key objectives of the project are:

- Extract sales, inventory, supplier, and purchase-order data from CSV datasets.
- Explore the datasets using Python and Pandas.
- Profile record counts, data types, missing values, duplicates, and identifiers.
- Validate product, store, supplier, and purchase-order relationships.
- Clean and transform the source datasets.
- Analyze product and store-level sales performance.
- Identify fast-moving and slow-moving products.
- Identify products and locations with inventory below reorder levels.
- Analyze supplier lead-time and order-frequency patterns.
- Analyze purchase-order quantities and actual lead times.
- Forecast next-month product demand.
- Calculate safety stock and reorder points.
- Perform supplier clustering using K-Means.
- Compare current inventory with calculated reorder points.
- Load the transformed data into MySQL.
- Build a structured MySQL Data Warehouse using fact and dimension tables.
- Perform SQL-based business analysis.
- Generate business insights related to demand, inventory, suppliers, and replenishment.

II. Architecture Overview

2.1 Architecture Overview

The Supply Chain & Inventory Management solution follows a structured **ETL, Data Warehouse, and Analytics architecture**.

The process begins with sales, inventory, supplier, and purchase-order datasets and progresses through data exploration, validation, cleaning, transformation, analytical processing, MySQL data warehousing, SQL analysis, and business insight generation.

The major components are:

1. Source Data

The project uses four CSV datasets:

- sales_data-2.csv
- inventory_data.csv
- suppliers_data.csv
- purchase_orders_data.csv

The datasets contain sales transactions, inventory information, supplier details, and purchase-order information.

2. Python / Google Colab

Python and Pandas were used for:

- Data loading
- Data exploration
- Data profiling
- Missing-value analysis
- Duplicate analysis
- Data-type validation
- Primary-key validation
- Relationship validation
- Data cleaning
- Data transformation
- Statistical analysis
- Demand analysis
- Forecasting

- Reorder-point calculation
- Supplier clustering

3. ETL & Cleaned Data

The source datasets were cleaned, validated, transformed, and prepared for analysis and warehouse loading.

Major processing activities included:

- Date conversion
- Missing-value validation
- Duplicate validation
- Product validation
- Store validation
- Supplier validation
- Purchase-order relationship validation
- Demand aggregation
- Inventory risk calculation
- Supplier feature preparation
- Forecasting dataset preparation

4. MySQL Data Warehouse

The cleaned data was loaded into the MySQL database:

supply_chain_inventory

The warehouse contains:

- dim_date
- dim_product
- dim_store
- dim_supplier
- fact_sales
- fact_inventory
- fact_purchase_orders

5. SQL Business Analysis

SQL was used to perform:

- Fast-moving and slow-moving product analysis
- Inventory reorder-risk analysis

- Store-level inventory risk analysis
- Product-store inventory analysis
- Supplier lead-time analysis
- Purchase-order analysis
- Sales trend analysis
- Sales and inventory comparison

6. Business Insights

The final analysis provides insights related to:

- Product demand
- Inventory risk
- Reorder requirements
- Forecasted demand
- Supplier characteristics
- Purchase-order performance
- Inventory optimization

No Power BI dashboard or DAX analysis was performed in this case study.

2.2 Architecture Flow



III. PROJECT APPROACH

3.1 Methodology

Source Data Identification

Four source datasets were identified:

- Sales Data
- Inventory Data
- Supplier Data
- Purchase Orders

The datasets were reviewed to understand their structure, fields, identifiers, and business meaning.

Initial dataset sizes:

Dataset	Records
Sales Data	5,000
Inventory Data	200
Supplier Data	50
Purchase Orders	1,000

Data Extraction & Exploration

The datasets were loaded into Google Colab using Python and Pandas.

The datasets were profiled using:

- Record counts
- Column names
- Data types
- Missing-value analysis
- Duplicate analysis
- Unique-ID analysis
- Numerical statistics
- Categorical analysis
- Date analysis
- Product analysis
- Store analysis
- Supplier analysis

Data Validation

Validation checks were performed for:

- Missing values
- Duplicate records
- Primary keys
- Product IDs
- Store IDs
- Supplier IDs
- Purchase-order relationships
- Date fields
- Numerical fields
- Categorical fields
- Data consistency

All four datasets were validated with **0 missing values and 0 duplicate records**.

Data Transformation

Data transformation included:

- Converting date columns into proper date formats
- Creating monthly sales summaries
- Creating product-level sales summaries
- Creating inventory-risk measures
- Calculating actual purchase-order lead time
- Preparing supplier clustering features
- Preparing monthly demand data
- Creating forecasting datasets
- Calculating reorder points and safety stock

Demand Forecasting

Historical monthly product demand was used for forecasting.

Three forecasting approaches were evaluated:

- 3-month Moving Average
- 6-month Moving Average
- 12-month Seasonal Naive

The model evaluation showed that the moving-average models performed better than the seasonal-naive approach on the test dataset, although forecast errors remained relatively high. Therefore, the forecasts were treated as planning estimates rather than exact demand predictions.

Reorder Point Calculation

Reorder points were calculated using:

- Average monthly demand
- Daily demand
- Demand variability
- Supplier lead time
- Safety stock
- 95% service-level assumption

Supplier Clustering

Supplier characteristics were prepared using:

- Supplier lead time
- Order frequency
- Number of purchase orders
- Total quantity ordered
- Average order quantity

K-Means clustering was evaluated using different cluster counts, with K = 3 selected for the final exploratory segmentation.

MySQL Data Warehouse

The cleaned data was loaded into MySQL.

A star-schema warehouse was created containing:

Dimension Tables

- dim_date
- dim_product
- dim_store
- dim_supplier

Fact Tables

- fact_sales
- fact_inventory
- fact_purchase_orders

SQL Business Analysis

SQL queries were then used to analyze:

- Product movement
- Inventory risk
- Store inventory risk
- Supplier lead time

- Purchase-order performance
- Sales trends

3.2 Technology Stack

Component	Technology	Purpose
Source Data	CSV	Sales, inventory, supplier and PO data
Data Processing	Python, Pandas	Exploration, cleaning and transformation
Development Environment	Google Colab	Python-based processing and analysis
Forecasting	Python	Demand forecasting
Clustering	Python / K-Means	Supplier segmentation
Data Warehouse	MySQL	Structured analytical storage
Database Tool	MySQL Workbench	Database creation, ETL and SQL analysis
Business Analysis	SQL	Inventory and supply-chain analysis
Documentation	MS Word	Project documentation
Component	Technology	Purpose
Source Data	CSV	Sales, inventory, supplier and PO data
Data Processing	Python, Pandas	Exploration, cleaning and transformation

IV. SOURCE DATA & TABLE MAPPING

4.1 Source Data

The project uses two primary source datasets.

- **Sales Dataset :** sales_data-2.csv

Records: 5,000

```
...
=====
SALES DATASET
=====
Rows      : 5000
Columns   : 6

Columns:
['Sale_ID', 'Product_ID', 'Store_ID', 'Sale_Date', 'Quantity_Sold', 'Revenue']

First 5 rows:
      Sale_ID Product_ID Store_ID Sale_Date Quantity_Sold Revenue
0  0d12b607      P001      S102  2024-01-01           322    2556
1  2e13fe47      P003      S101  2024-01-02            28   42606
2  c05afd2d      P003      S104  2024-01-03           292   19551
3  aff62361      P004      S101  2024-01-04           280   26625
4  9d190af1      P003      S101  2024-01-05           353   24001
```

- **Inventory Dataset :** inventory_data.csv

Records: 200

```
=====
INVENTORY DATASET
=====
Rows      : 200
Columns   : 6

Columns:
['Product_ID', 'Store_ID', 'Warehouse_ID', 'Stock_Level', 'Reorder_Level', 'Last_Updated']

First 5 rows:
      Product_ID Store_ID Warehouse_ID Stock_Level Reorder_Level Last_Updated
0          P005      S101          W003          498           163    2024-02-01
1          P004      S103          W003            24           178    2024-02-02
2          P002      S101          W002          245           188    2024-02-03
3          P003      S101          W003            61           148    2024-02-04
4          P002      S101          W002            57           155    2024-02-05
```

- **Supplier Dataset:** suppliers_data.csv

Records: 50

```
=====
SUPPLIERS DATASET
=====
Rows      : 50
Columns   : 5

Columns:
['Supplier_ID', 'Supplier_Name', 'Product_ID', 'Lead_Time (days)', 'Order_Frequency']

First 5 rows:
```

	Supplier_ID	Supplier_Name	Product_ID	Lead_Time (days)	Order_Frequency
0	SUP001	Murray-Ramirez	P002	9	Weekly
1	SUP002	Thomas, Meyer and Campbell	P005	9	Weekly
2	SUP003	Acosta Inc	P005	4	Monthly
3	SUP004	Evans, Brown and Turner	P002	8	Weekly
4	SUP005	Johnson, Wise and Barr	P003	6	Biweekly

- **Purchase Orders Dataset :** purchase_orders_data.csv

Records: 1,000

```
=====
PURCHASE ORDERS DATASET
=====
Rows      : 1000
Columns   : 6

Columns:
['Order_ID', 'Product_ID', 'Supplier_ID', 'Order_Date', 'Quantity', 'Arrival_Date']

First 5 rows:
```

	Order_ID	Product_ID	Supplier_ID	Order_Date	Quantity	Arrival_Date
0	PO0001	P001	SUP002	2024-01-01	158	2024-01-06
1	PO0002	P005	SUP019	2024-01-02	440	2024-01-07
2	PO0003	P002	SUP036	2024-01-03	423	2024-01-08
3	PO0004	P004	SUP021	2024-01-04	88	2024-01-09
4	PO0005	P001	SUP002	2024-01-05	329	2024-01-10

V. DATA EXPLORATION & PRIMARY KEY VALIDATION

5.1 Data Exploration

Initial data profiling was performed using Python and Pandas.

The analysis included:

- Record counts
- Column names
- Data types
- Missing-value analysis
- Duplicate analysis
- Primary-key validation
- Product analysis
- Store analysis
- Inventory analysis
- Supplier analysis
- Purchase-order analysis
- Date validation

Dataset	Records
Sales Data	5,000
Purchase Orders	1000
Inventory Data	200
Supplier Data	50

5.2 Data Quality Validation

All four datasets contained:

- **0 missing values**
- **0 duplicate records**

Primary-key validation also confirmed unique identifiers for the applicable datasets.

Relationship validation confirmed:

- Product IDs were consistent.
- Store IDs were consistent.
- Supplier IDs in purchase orders existed in the supplier dataset.
- Date fields were successfully converted.
- No invalid purchase-order date relationships were identified.

5.3 Sales Analysis

The sales dataset contained:

- **5,000 transactions**
- **1,292,122 units sold**
- **125,870,014 total revenue**

The average quantity per transaction was approximately **258.42 units**.

Product-level analysis showed that:

- P002 recorded the highest quantity sold at **269,100 units**.
- P001 recorded **260,296 units**.
- P004 recorded **258,635 units**.
- P005 recorded **254,794 units**.
- P003 recorded the lowest quantity sold at **249,297 units**.

5.4 Inventory Validation

The inventory dataset contained:

- **200 inventory records**
- **44,991 total stock**
- **24,958 total reorder level**

Inventory status analysis identified:

- **53 records below reorder level**
- **1 record at reorder level**
- **146 records above reorder level**

The overall record-level reorder risk was:

26.5%

5.5 Supplier & Purchase Order Validation

The supplier dataset contained 50 suppliers.

Supplier lead times ranged from:

3 to 9 days

Purchase-order analysis contained:

- 1,000 purchase orders
- 276,488 total quantity ordered
- 5 days average actual lead time

The actual purchase-order lead time was consistently 5 days across all purchase orders.

VI. MYSQL DATA WAREHOUSE IMPLEMENTATION & DATA LOADING

6.1 MySQL DataBase Implementation

After data cleaning and transformation, the analytical data was loaded into:

supply_chain_inventory

MySQL was used to provide a structured relational environment for storing supply-chain dimensions and analytical facts.



Figure 6.1: Tables created in MySQL Workbench

6.2 Data Warehouse Structure

- Inventory_data

	Product_ID	Store_ID	Warehouse_ID	Stock_Level	Reorder_Level	Last_Updated
▶	P005	S101	W003	498	163	2024-02-01
	P004	S103	W003	24	178	2024-02-02
	P002	S101	W002	245	188	2024-02-03
	P003	S101	W003	61	148	2024-02-04
	P002	S101	W002	57	155	2024-02-05
	P003	S103	W002	443	63	2024-02-06
	P005	S102	W002	392	105	2024-02-07

- Purchase_orders

	Order_ID	Product_ID	Supplier_ID	Order_Date	Quantity	Arrival_Date
▶	PO0001	P001	SUP002	2024-01-01	158	2024-01-06
	PO0002	P005	SUP019	2024-01-02	440	2024-01-07
	PO0003	P002	SUP036	2024-01-03	423	2024-01-08
	PO0004	P004	SUP021	2024-01-04	88	2024-01-09
	PO0005	P001	SUP002	2024-01-05	329	2024-01-10

- Sales_data

	Sale_ID	Product_ID	Store_ID	Sale_Date	Quantity_Sold	Revenue
▶	000d012c	P001	S103	2030-09-26	31	30216.00
	0010d204	P002	S101	2026-03-03	480	40447.00
	00165c7a	P004	S102	2024-04-19	233	47809.00
	0016b318	P003	S103	2032-08-11	374	1066.00
	00279b37	P002	S103	2036-07-14	437	27786.00
	002f5a10	P001	S103	2026-01-22	281	35884.00
	003c59f7	P003	S104	2028-08-16	183	19478.00

- Suppliers_data

	Supplier_ID	Supplier_Name	Product_ID	Lead_Time (days)	Order_Frequency
▶	SUP001	Murray-Ramirez	P002	9	Weekly
	SUP002	Thomas, Meyer and Campbell	P005	9	Weekly
	SUP003	Acosta Inc	P005	4	Monthly
	SUP004	Evans, Brown and Turner	P002	8	Weekly
	SUP005	Johnson, Wise and Barr	P003	6	Biweekly
	SUP006	Hood, Barrera and Woodard	P001	6	Monthly
	SUP007	Meyers, Miller and Young	P002	9	Weekly

VII. SQL BUSINESS ANALYSIS

7.1 Product Movement Analysis

SQL was used to classify products into fast-moving and slow-moving categories based on product-level sales quantity.

Fast-moving products

- P001
- P002
- P004

Slow-moving products

- P003
- P005

P002 recorded the highest quantity sold among the five products.

7.2 Inventory Reorder Risk Analysis

SQL analysis identified products with inventory records below their reorder levels.

Product	Inventory Records	Below Reorder	Risk
P001	40	10	25.00%
P002	39	8	20.51%
P003	51	14	27.45%
P004	36	12	33.33%
P005	34	9	26.47%

The analysis indicates that inventory risk varies by product and store location.

7.3 Store Inventory Risk Analysis

Store-level analysis showed:

Store	Records	Below Reorder	Risk
S101	59	19	32.20%
S102	48	16	33.33%
S103	53	9	16.98%
S104	40	9	22.50%

This analysis helps identify stores where inventory monitoring may require greater attention.

7.4 Supplier Lead-Time Analysis

Supplier master lead times ranged from **3 to 9 days**.

Product-level supplier analysis showed:

- P001: average lead time approximately **6.39 days**
- P002: approximately **6.69 days**
- P003: approximately **6.12 days**
- P004: approximately **5.55 days**
- P005: approximately **6.05 days**

These values represent the **supplier master lead-time information**, not actual PO delivery performance.

7.5 Purchase Order Performance

Actual purchase-order analysis showed:

- P001: 194 POs
- P002: 216 POs
- P003: 206 POs
- P004: 198 POs
- P005: 186 POs

Actual lead time for all purchase orders was **5 days**.

Therefore, the dataset shows **no variation in actual PO lead time**, even though supplier master lead times vary from 3 to 9 days.

7.6 Sales Trend Analysis

Monthly sales analysis covered the period from **January 2024 to September 2037**.

September 2037 was identified as a **partial month**, containing only 8 days of sales data.

Therefore, September 2037 was not treated as a complete monthly demand period during forecasting preparation.

VIII. ADVANCED ANALYTICS

8.1 Demand Forecasting

Historical monthly product demand was prepared for forecasting.

Three approaches were evaluated:

- 3-Month Moving Average
- 6-Month Moving Average
- 12-Month Seasonal Naive

Overall model evaluation showed:

Model	MAE	RMSE	MAPE
3-Month Moving Average	556.93	666.12	41.21%
6-Month Moving Average	545.03	663.74	40.66%
12-Month Seasonal Naive	792.65	990.99	57.52%

The moving-average models produced lower errors than the seasonal-naive model.

The selected product-specific forecasting approaches generated the following September 2037 demand estimates:

Product	Forecast Units
P001	1,170
P002	1,373
P003	1,345
P004	1,852
P005	1,466
Total	7,206

8.2 Reorder Point Analysis

Reorder points were calculated using demand, demand variability, supplier lead time, and safety stock.

Product	Calculated ROP
P001	876.72
P002	904.05
P003	766.86
P004	817.45
P005	805.81

The calculated reorder points provide a demand-based reference for replenishment planning.

8.3 Supplier Clustering

Supplier clustering was performed using K-Means based on supplier characteristics including:

- Lead time
- Order frequency
- Purchase-order count
- Total quantity ordered
- Average order quantity

Three clusters were selected for exploratory supplier segmentation.

The clusters showed different combinations of supplier lead time, order frequency, purchase-order volume, and order quantity.

The clustering is intended to support **supplier segmentation and comparative analysis**, rather than declaring a supplier group as inherently better or worse.

IX. BUSINESS ANALYTICAL INSIGHTS

9.1 Sales & Demand Insights

The sales analysis shows that **P002 recorded the highest total quantity sold at 269,100 units**, while P003 recorded the lowest at 249,297 units.

P001, P002, and P004 were classified as fast-moving products based on the defined quantity-based classification.

This indicates that product-level demand varies and should be considered when planning inventory replenishment.

9.2 Inventory Risk Insights

The overall inventory analysis identified **53 out of 200 inventory records below their reorder level**, representing a **26.5% record-level reorder risk**.

P004 had the highest product-level reorder risk at **33.33%**, while P002 had the lowest at **20.51%**.

This indicates that inventory risk is not uniform across products.

9.3 Store-Level Inventory Insights

S102 showed a reorder risk of **33.33%**, while S101 showed **32.20%**.

S103 recorded the lowest reorder risk at **16.98%**.

This indicates that inventory conditions differ between stores and that store-level inventory monitoring can help identify locations requiring closer replenishment attention.

9.4 Demand and Inventory Insight

P004 combines:

- High sales activity
- 258,635 units sold
- Fast-moving classification
- 33.33% inventory reorder risk

This makes P004 an important product for monitoring because demand activity and inventory risk occur together.

P002 recorded the highest sales quantity while having a lower product-level reorder risk of **20.51%**.

9.5 Forecasting Insight

The forecast for the next complete planning month produced an estimated total demand of **7,206 units** across the five products.

P004 had the highest forecast at **1,852 units**, followed by P005 at **1,466 units**, P002 at **1,373 units**, P003 at **1,345 units**, and P001 at **1,170 units**.

These forecasts can be used as planning estimates for inventory and replenishment analysis.

9.6 Reorder Point Insight

Calculated reorder points ranged from:

- **766.86 units for P003**
- **to 904.05 units for P002**

The calculated ROP values provide a demand- and variability-based reference for replenishment planning.

The comparison between existing reorder levels and calculated ROP shows substantial differences. However, this difference should not automatically be interpreted as an error in the original reorder levels because the two measures are based on different assumptions and methodologies.

9.7 Supplier Insight

Supplier master lead times vary between **3 and 9 days**, while actual purchase-order lead time is consistently **5 days**.

This difference indicates that supplier master lead-time information and observed PO delivery performance represent different measures and should be analyzed separately.

Supplier clustering additionally provides a structured way to segment suppliers according to lead time, order frequency, and purchasing activity.

9.8 Inventory Optimization Insight

The optimization analysis compared current product-level stock with calculated ROP and forecast demand.

Current aggregate stock across products was:

44,991 units

After subtracting the total next-month forecast of **7,206 units**, the modeled remaining stock was:

37,785 units

At the product-level aggregate comparison, current stock remained above the calculated ROP for all five products.

However, this should be interpreted as a **model-based planning comparison**, because the inventory data is an earlier snapshot while the forecasting period extends much later.

X. CONCLUSION

11.1 Conclusion

The **Supply Chain & Inventory Management** case study implemented a structured analytics solution using **Python, Google Colab, MySQL, and SQL**.

The sales, inventory, supplier, and purchase-order datasets were extracted, explored, validated, cleaned, transformed, and analyzed. Python was used for data profiling, inventory analysis, demand forecasting, reorder-point calculation, and supplier clustering.

A structured **MySQL Data Warehouse** was then created using dimension and fact tables. SQL analysis was performed to identify product movement, inventory reorder risk, store-level risk, supplier lead-time patterns, purchase-order performance, and sales trends.

The analysis provides a structured view of demand, inventory conditions, supplier characteristics, and replenishment requirements. The calculated forecasts and reorder points provide additional planning references for inventory management.

The project was completed up to the **MySQL Workbench and SQL Business Analysis stage**. **Power BI was not implemented** as part of this case study.

11.2 Key Outcomes

- Implemented an end-to-end ETL and analytics pipeline using Python and MySQL.
- Profiled **5,000 sales records**.
- Profiled **200 inventory records**.
- Profiled **50 supplier records**.
- Profiled **1,000 purchase orders**.
- Validated all source datasets for missing values and duplicates.
- Identified fast-moving and slow-moving products.
- Identified **53 inventory records below reorder level**.
- Calculated an overall record-level reorder risk of **26.5%**.
- Analyzed product-level and store-level inventory risk.
- Analyzed supplier lead-time and order-frequency patterns.
- Validated **1,000 purchase orders** with an actual lead time of **5 days**.

- Evaluated three demand forecasting approaches.
- Generated next-month demand forecasts totaling **7,206 units**.
- Calculated product-level safety stock and reorder points.
- Performed supplier clustering using K-Means.
- Created a MySQL star-schema data warehouse.
- Created **4 dimension tables** and **3 fact tables**.
- Validated all fact-to-dimension relationships with **0 invalid records**.
- Reconciled source and warehouse totals successfully.
- Performed SQL-based supply-chain and inventory business analysis.
- Generated business insights for demand, inventory, replenishment, and supplier management.
- Completed the project up to the **MySQL Workbench / SQL analysis stage**.

11.2 Github Link

<https://github.com/surabhi-0405/CaseStudy-6-Supply-Chain-Inventory-Management>